Computer Graphics

Week 1 : *Introduction*

christianskaditya@umm.ac.id

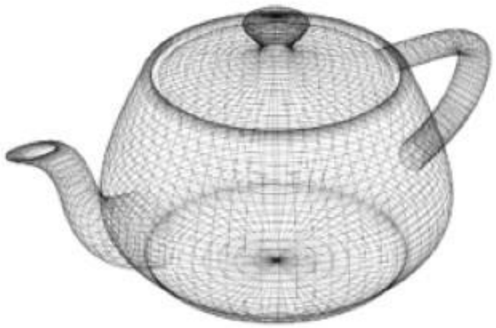
OUTLINES

1. Understanding Computer Graphics
2. Type of Digital Files (Vector vs Raster)

1. Understanding Computer Graphics

Computer graphics are pictures created using computers.

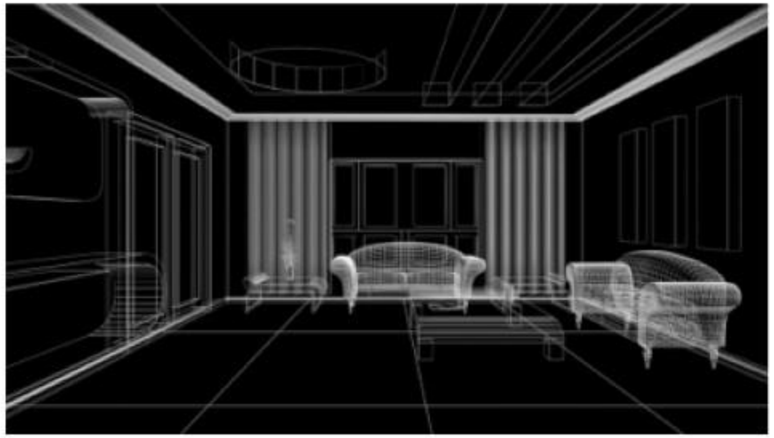
Usually, computer-generated image data created with the help of specialized graphical hardware and software.



Computer graphics development has had a significant impact on many types of media and has revolutionized animation, movies, advertising, video games, and graphic design in general.

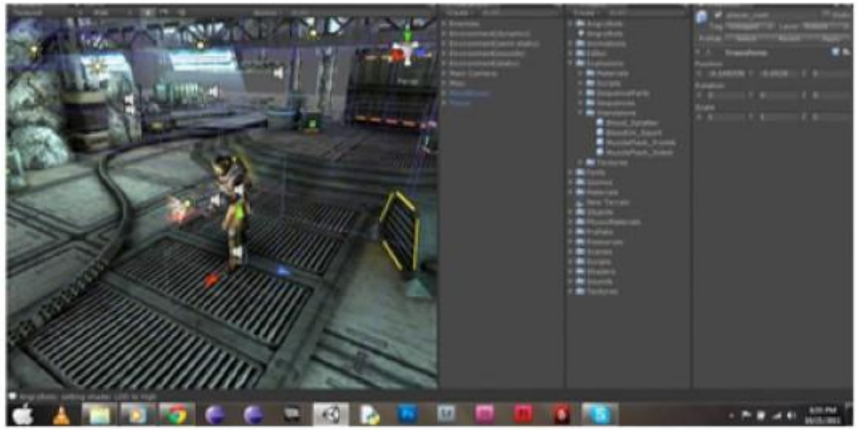
The following are specific application fields of computer graphics:

1. Data Visualization



3. Design

2. Entertainment



4. Game Development

Computer Graphic Element

- Shape
- Position
- Orientation
- Surface Properties
- Volumetric Properties
- Lights

christianskaditya@umm.ac.id

Computer Graphics relates to:

a) **Geometric Modeling**: creating mathematical models of 2D and 3D objects.
b) **Rendering**: produce images given to this model.
c) **Animation**: define / display the behavior / behavior of objects depending on the time.

1a. **Geometric Modeling**

Geometric modelling is the process of capturing the properties of an object or a system using **mathematical formulas**.

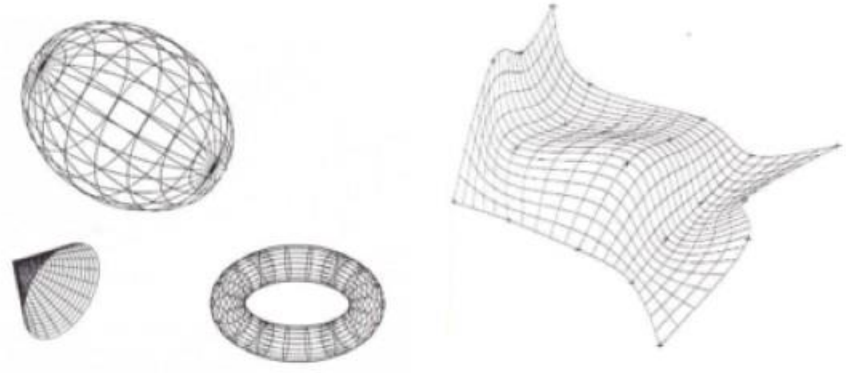
Computer geometric modelling is the field that discusses the mathematical methods behind the modelling of realistic objects for computer graphics and computer aided design.

It uses computers to store the data and the geometric properties of a image.

A geometric model contains description of the **modelled object's shape**. Since geometric shapes are described by surfaces, curves are used to construct them.

Computer geometric modelling uses curves to control the object's surfaces as they are easy to manipulate. The curves may be constructed using analytic functions, a set of points, or other curves and surfaces.



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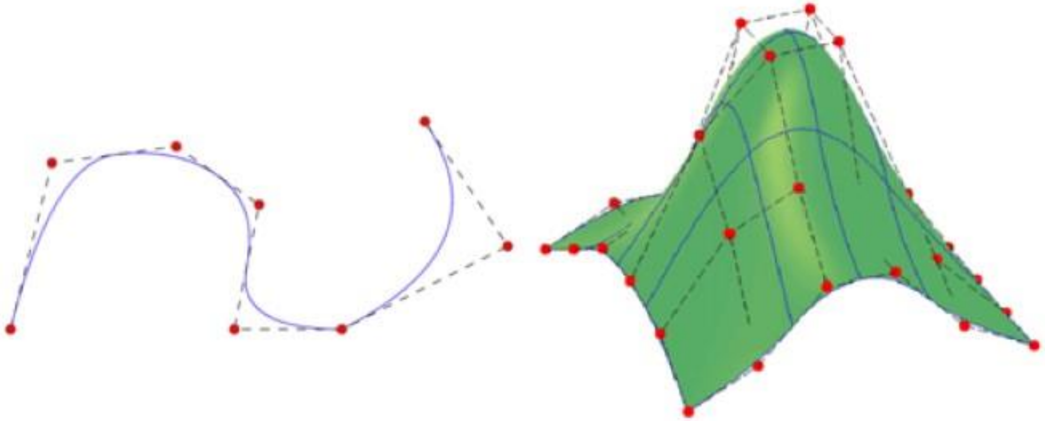
A geometric model of an object can be created using these 3 steps:

- Create basic geometric object using the commands like points, line and circles in Computer Aided Design (CAD) software
- Use commands like achieve scaling, rotation, etc. to transform these geometric elements
- Integrate the various elements of the object to form the final geometric model.



1b. **Rendering**

2. Rendering

Rendering or image synthesis is the process of generating a final digital product from a specific type of input (from a 2D or 3D model). The term usually applies to **graphics** and **video**, but it can refer to **audio** as well.

3D graphics are rendered from basic three-dimensional models called wireframes.

A wireframe defines the shape of the model, but nothing else. The rendering process adds surfaces, textures, and lighting to the model, giving it a realistic appearance.

Central Processing Unit (CPU) is often optimized to run multiple smaller tasks at the same time.

Graphic Processing Unit (GPU) is a purpose-built device able to assist a CPU in performing complex rendering calculations.

GPU have several dozens or several hundred cores more than CPU. But it doesn't mean that they're just faster than CPU. Both units are designed for different goals and are very different.

In a simplified way, **CPU has to perform complicated operations, using a small amount of data and GPU has to carry out simple operations, but using a lot of data.**

GPU is **not** faster than the CPU. CPU and GPU are designed with two different goals, with different trade-offs, so they have **different** performance characteristic. Certain tasks are faster in a CPU while other tasks are faster computed in a GPU. The CPU excels at doing complex manipulations to a small set of data, the **GPU excels at doing simple manipulations to a large set of data.**

PS : **Arithmetic-Logic Unit (ALU)** --> serve as a combinational digital circuit that performs arithmetic and bitwise operations on binary numbers.

1c. **Computer Animation**

3. Animation

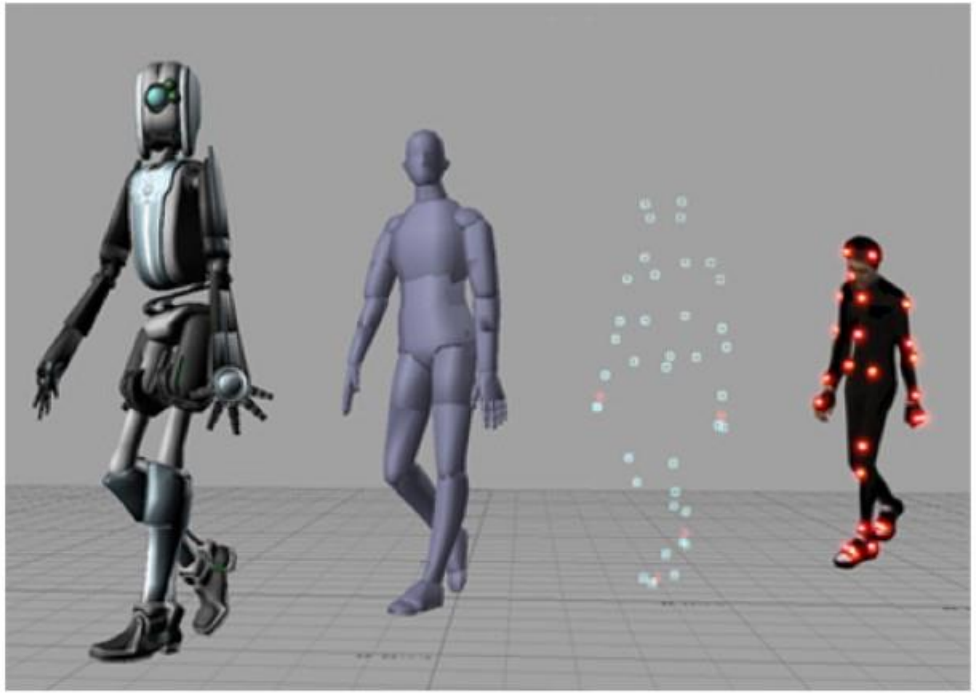
Animation means giving life to any object in computer graphics. It has the power of injecting energy and emotions into the most seemingly inanimate objects.

The basic idea behind animation is to **play back the recorded images** at the rates fast enough to fool the human eye into interpreting them as continuous motion.

Animation can make a series of dead images come alive. Animation can be used in many areas like entertainment, computer aided-design, scientific visualization, training, education, e-commerce, and computer art.

For 3D animations, objects (models) are built on the computer monitor (modeled) and 3D figures are rigged with a virtual skeleton.

For 2D figure animations, separate objects (illustrations) and separate transparent layers are used, with or without a virtual skeleton.

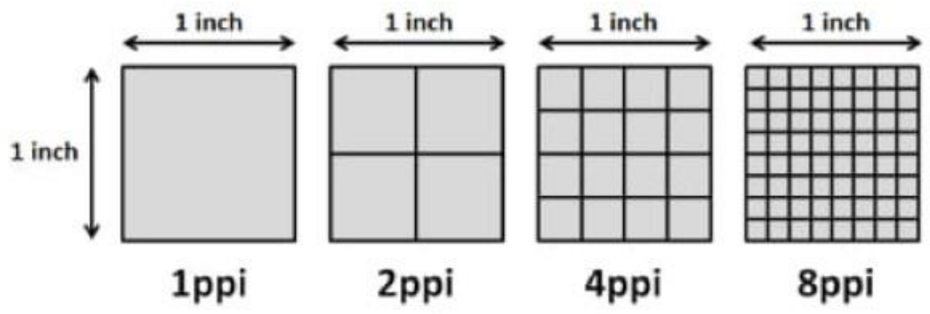


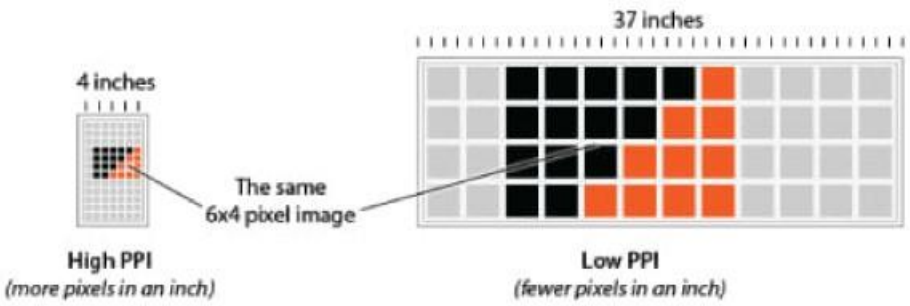
2. Type of Digital Images (Raster vs Vector)

Raster

Raster graphics also known as bitmaps are composed of pixels. such as a “.gif” or “.jpeg”, is an array of pixels of various colors, which together form an image.

Digital cameras create raster images, and all the photographs you see in print and online are raster images. Quality is often dictated by how many pixels are contained in an inch, expressed as pixels-per-inch or ppi.

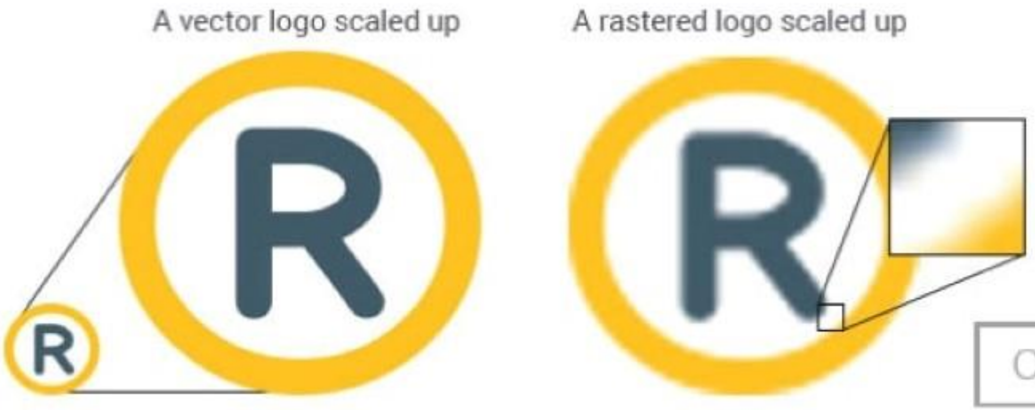


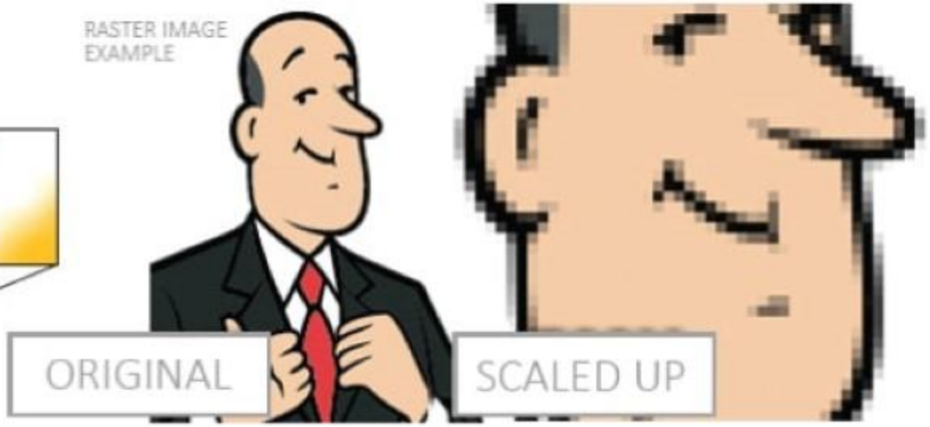


Raster (2)

The greater the ppi (pixels per inch) and dimensional measurements, the higher the quality.

When they are scaled (zoom in), quality is lost and they become blurry. which is typically the case for web images, often saved in smaller sizes and at resolutions of either 72ppi or 96ppi.

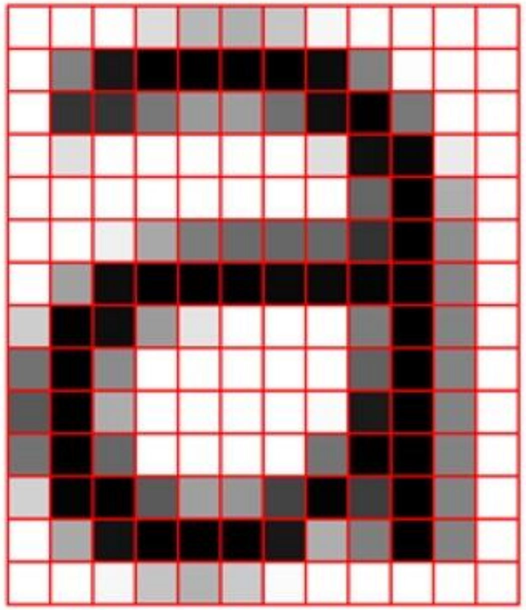




Raster (3)

Raster (bitmap) images use bit maps to store information.

a



1.0	1.0	1.0	0.9	0.6	0.6	0.6	1.0	1.0	1.0	1.0	1.0
1.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.5	1.0	1.0	1.0
1.0	0.2	0.2	0.5	0.6	0.6	0.5	0.0	0.0	0.5	1.0	1.0
1.0	0.9	1.0	1.0	1.0	1.0	1.0	0.9	0.0	0.0	0.9	1.0
1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.5	0.0	0.5	1.0
1.0	1.0	1.0	0.5	0.5	0.5	0.5	0.5	0.4	0.0	0.5	1.0
1.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	1.0
0.9	0.0	0.0	0.6	1.0	1.0	1.0	1.0	0.5	0.0	0.5	1.0
0.5	0.0	0.6	1.0	1.0	1.0	1.0	1.0	0.5	0.0	0.5	1.0
0.5	0.0	0.7	1.0	1.0	1.0	1.0	1.0	0.0	0.0	0.5	1.0
0.6	0.0	0.6	1.0	1.0	1.0	1.0	0.5	0.0	0.0	0.5	1.0
0.9	0.1	0.0	0.6	0.7	0.7	0.5	0.0	0.5	0.0	0.5	1.0
1.0	0.7	0.1	0.0	0.0	0.0	0.1	0.9	0.8	0.0	0.5	1.0
1.0	1.0	1.0	0.8	0.8	0.9	1.0	1.0	1.0	1.0	1.0	1.0

Vector

Vector graphics are composed of paths. A vector graphic, such as an “.eps” file (corel draw) or “.ai” file (adobe illustrator), is composed of paths, or lines, that are either straight or curved.

The data file for a vector image contains the points where the paths start and end, how much the paths curve, and the colors that either border or fill the paths.

Because vector graphics are not made of pixels, the images can be scaled to be very large without losing quality. Raster graphics, on the other hand, become "blocky," since each pixel increases in size as the image is made larger.

Vector (2)

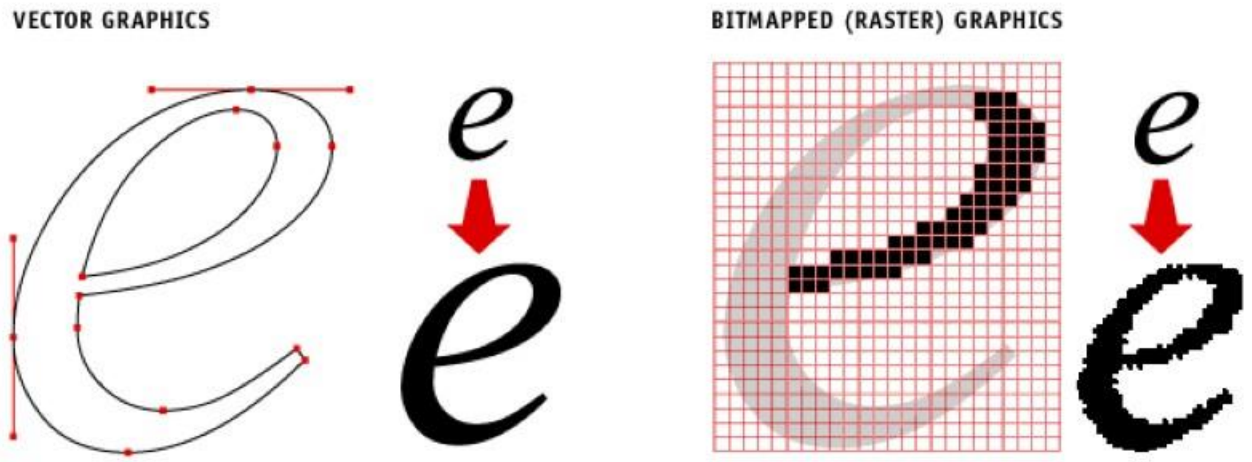
Since mathematical formulas dictate how the image is rendered, vector images retain their appearance regardless of size. They can be scaled infinitely.

Vector images are comprised of shapes, and each shape has its own color; thus, vectors cannot achieve the color gradients, shadows, and shading that raster images can (it is possible to mimic them, but it requires rasterizing part of the image – which means it would not be a true vector).

Vector (3)

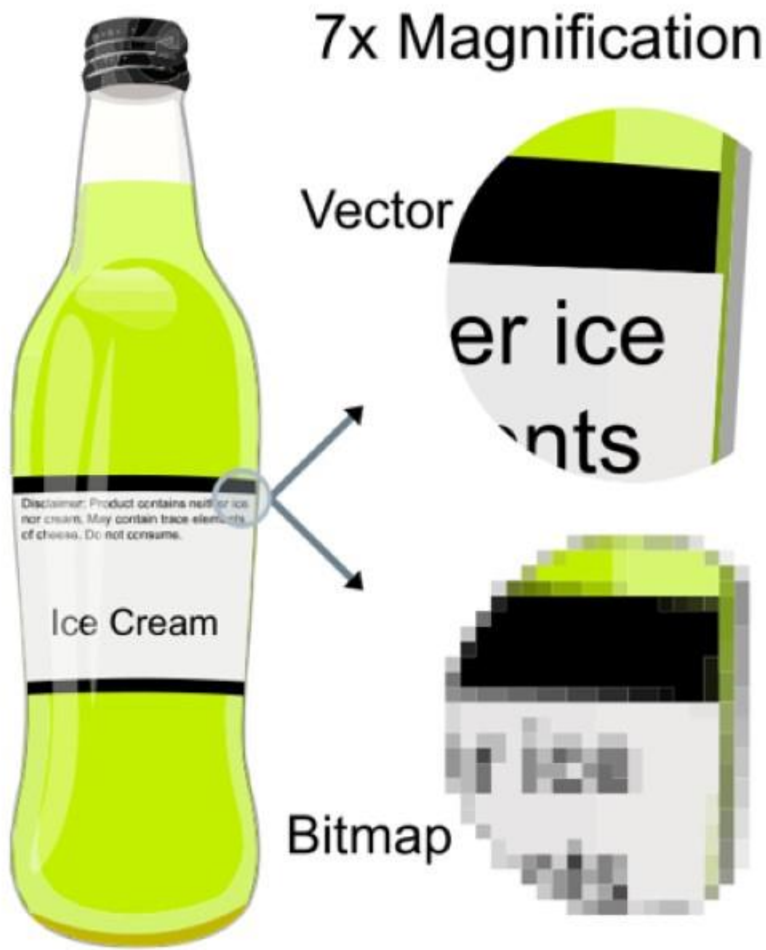
Vector graphics are best for printing since it is composed of a series of mathematical curves. As a result vector graphics print crisply even when they are enlarged.

How are raster images and vector graphics different?



Pixels vs Math

Raster images are comprised of colored pixels arranged to form an image, they cannot be scaled without sacrificing quality. In contrast, the mathematical equations that form the foundations for vectors recalculate when they're resized; thus, you can infinitely scale a vector graphic


7x Magnification
Vector
er ice
Bitmap
Ice Cream

True-to-Life Graphics

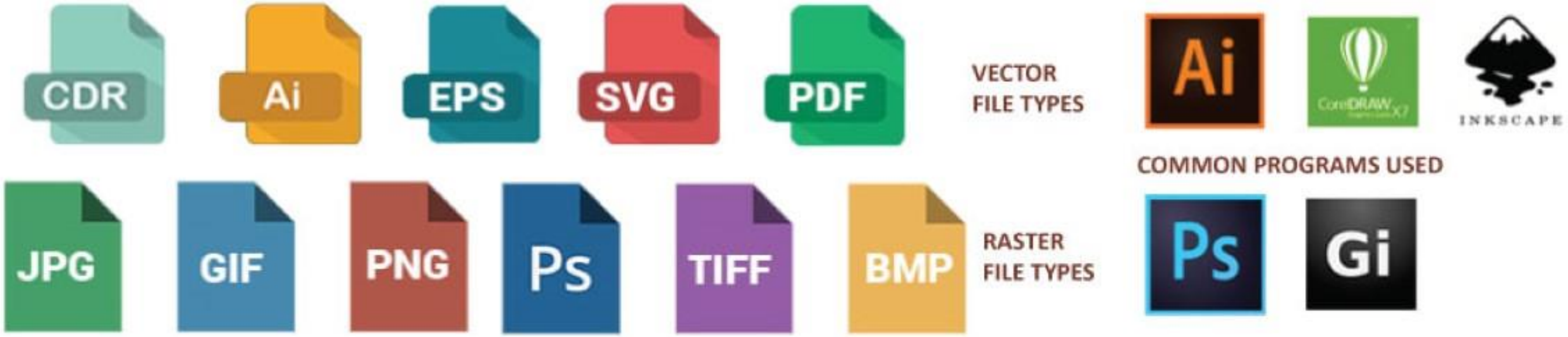
Though it’s possible to make a vector resemble a photograph, the minute nuances of blended colors, shading, shadows, and gradient make it impossible to get a true-to-life representation of a photograph with vectors.

Rasterized images, on the other hand, are perfectly capable of rendering true-to-life graphics: visually-perfect color blends, shades, gradients, and shadows. Of course, unlike vectors, they’re still limited by dimensional size and resolution.

File Extension

The most common raster file types include JPG, GIF, PNG, TIF, BMP, and PSD. The most common vector file types are AI, CDR, EPS, and SVG.

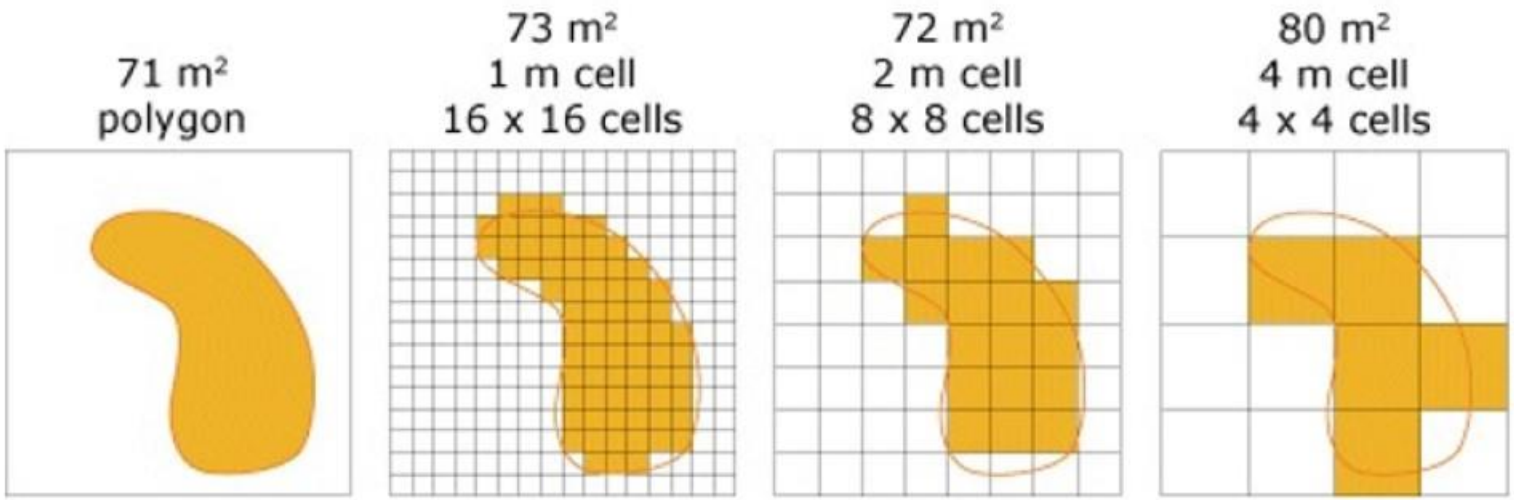
Both rasters and vectors can be rendered in EPS and PDF format, where the software that created the file dictates whether it's a raster or vector file.



File Size

Compared to vectors, rasters take up a lot of space. Why? Since vectors rely on calculations to be performed by the programs that load them, the only information they need to contain are their mathematical formulas.

Raster images use bit maps to store information. This means a large file needs a large bitmap. The larger the image, the more disk space the image file will take up. As an example, a 640 x 480 image requires information to be stored for 307,200 pixels, while a 3072 x 2048 image needs to store information for a whopping 6,291,456 pixels or 6.3 Megapixel .



- **Smaller** cell size
- **Higher** resolution
- **Higher** accuracy
- **Slower** display
- **Slower** processing
- **Larger** file size

- **Larger** cell size
- **Lower** resolution
- **Lower** accuracy
- **Faster** display
- **Faster** processing
- **Smaller** file size

Conversion : Vector to Raster

Printers and display devices are raster devices. As a result we need to convert vector images to raster format before they can be used i.e displayed or printed. The required resolution plays an vital role in determining the size of raster file generated.

Here it is important to note that the size of vector image to be converted always remains the same. It is convenient to convert a vector file to a range of bitmap/raster file formats but going down opposite path is harder. (because at times we need to edit the image while converting from raster to vector)

Raster	Vector
Comprised of pixels, arranged to form an image	Comprised of paths, dictated by mathematical formulas
Constrained by resolution and dimensions	Infinitely scalable
Capable of rich, complex color blends	Difficult to blend colors without rasterizing
Large file sizes (but can be compressed)	Small file sizes
File types include .jpg, .gif, .png, .tif, .bmp, .psd; plus .eps and .pdf when created by raster programs	File types include .ai, .cdr, .svg; plus .eps and .pdf when created by vector programs
Raster software includes Photoshop and GIMP	Vector software includes Illustrator, CorelDraw, and InkScape
Perfect for "painting" / photos	Perfect for "drawing" / logos
Capable of detailed editing	Less detailed, but offers precise paths

PRINT SIZE	IMAGE DIMENSIONS (300dpi)	MEGAPIXELS NEEDED
4 x 6”	1200 x 1800 pixels	2.1 MP
5 x 7”	1500 x 2100 pixels	3.1 MP
8 x 10”	2400 x 3000 pixels	7.2 MP
8 x 12”	2400 x 3600 pixels	8.6 MP
11 x 14”	3300 x 4200 pixels	13.8 MP
16 x 24”	4800 x 7200 pixels	34.5 MP
20 x 24”	6000 x 7200 pixels	43.2 MP

THANKYOU